4.

35 17 10 7 28 23 41 15 20 29

(a) Use the first-fit bin packing algorithm to determine how the numbers listed above can be packed into bins of size 60

(3)

(b) The list of numbers is to be sorted into descending order. Use a quick sort to obtain the sorted list. You should show the result of each pass and identify your pivots clearly.

(4)

(c) Use the first-fit decreasing bin packing algorithm on your ordered list to pack the numbers into bins of size 60

(3)

The ten distinct numbers below are to be sorted into descending order.

20 24 17 26 8 15 x y 19 12

A bubble sort, starting at the left-hand end of the list, is to be used to obtain the sorted list.

After the second complete pass the list is

24 26 20 17 15 y 19 12 x 8

(d) Find the constraints on the values of x and y.

(3)

(Total 13 marks)

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4.

35 17 10 7 28 23 41 15 20 29

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total 13 marks)

Q4

Write your answers in the D1 answer book for this paper.

1. Use the binary search algorithm to try to locate the word "Parallelogram" in the following alphabetical list. Clearly indicate how you choose your pivots and which part of the list you are rejecting at each stage.

Arc

Centre

Chord

Circle

Circumference

Diameter

Radius

Sector

Segment

Tangent

(Total 4 marks)

Leave blank

1.

Q1

(Total 4 marks)

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3. 2.6 0.8 2.1 1.2 0.9 1.7 2.3 0.3 1.8 2.7

(a) Use the first-fit bin packing algorithm to determine how the numbers listed above can be packed into bins of size 5

(3)

The list is to be sorted into **descending** order.

(b) (i) Starting at the left-hand end of the above list, perform **two** passes through the list using a bubble sort. Write down the lists that result at the end of the first pass and the second pass.

(ii) Write down, in the table in the answer book, the number of comparisons and the number of swaps performed during each of these two passes.

(4)

After a third pass using this bubble sort, the updated list is

2.6 2.1 1.7 2.3 1.2 1.8 2.7 0.9 0.8 0.3

(c) Use a quick sort on this updated list to obtain the fully sorted list. You must make your pivots clear.

(3)

(d) Apply the first-fit decreasing bin packing algorithm to the fully sorted list to pack the numbers into bins of size 5

(3)

(Total 13 marks)

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Question 3 continued

	Number of comparisons	Number of swaps
First pass		
Second pass		

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Question 3 continued

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Question 3 continued

(Total 13 marks)

Q3

Write your answers in the D1 answer book for this paper.

1.

17 9 15 8 20 13 28 4 12 5

The numbers in the list shown above are the weights, in kilograms, of ten boxes.
The boxes are to be transported in containers that will each hold a maximum weight of 40 kilograms.

(a) Calculate a lower bound for the number of containers that will be needed to transport the boxes. You must show your working.

(2)

(b) Use the first-fit bin packing algorithm to allocate the boxes to the containers.

(3)

(c) Using the list provided, carry out a quick sort to produce a list of the weights in **ascending** order. You must make your pivots clear.

(3)

(d) Use the binary search algorithm to try to locate the weight of 9 in the sorted list. Clearly indicate how you choose your pivots and which part of the list is being rejected at each stage.

(3)

(Total 11 marks)

1.

17 9 15 8 20 13 28 4 12 5

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Question 1 continued

Question 1 continued

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Question 1 continued

(Total 11 marks)

3.

1.8 1.4 2.6 1.6 2.8 0.9 3.1 0.8 1.2 2.4 0.6

(a) Use the first-fit bin packing algorithm to determine how the numbers listed above can be packed into bins of size 5

(3)

The list is to be sorted into descending order.

(b) (i) Perform one pass of a bubble sort, starting at the left-hand end of the list. You must write down the list that results at the end of the first pass.

(ii) Write down the number of comparisons and the number of swaps performed during the first pass.

(3)

After a second pass using this bubble sort, the updated list is

2.6 1.8 2.8 1.6 3.1 1.4 1.2 2.4 0.9 0.8 0.6

(c) Use a quick sort on this updated list to obtain the fully sorted list in descending order. You must make your pivots clear.

(4)

(d) Apply the first-fit decreasing bin packing algorithm to the fully sorted list to pack the numbers into bins of size 5

(3)

(Total for Question 3 is 13 marks)

3.

1.8 1.4 2.6 1.6 2.8 0.9 3.1 0.8 1.2 2.4 0.6

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Question 3 continued

Question 3 continued

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Question 3 continued

(Total for Question 3 is 13 marks)

6. The twelve numbers in the list below are to be packed into bins of size n, where n is a positive integer.

28 31 5 25 16 35 18 22 11 27 15 13

When the first-fit bin packing algorithm is applied to the list, the following allocation is obtained.

Bin 1: 28 31 5

Bin 2: 25 16 18 11

Bin 3: 35 22 15

Bin 4: 27 13

(a) Based on the packing shown above, determine the possible values of n. You must give reasons for your answer.

(3)

(b) The original list of twelve numbers is to be sorted into **ascending** order. Use a quick sort to obtain the sorted list. You should show the result of each pass and identify your pivots clearly.

(4)

When the first-fit decreasing bin packing algorithm is applied to the list, the following allocation is obtained.

Bin 1: 35 31 5

Bin 2: 28 27 16

Bin 3: 25 22 18

Bin 4: 15 13 11

(c) Determine the value of n. You must give a reason for your answer.

(2)

(Total for Question 6 is 9 marks)

6.

28 31 5 25 16 35 18 22 11 27 15 13

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Question 6 continued

28 31 5 25 16 35 18 22 11 27 15 13

(Total for Question 6 is 9 marks)

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2. (a) (i) Describe how to carry out the first pass of a bubble sort when it is used to sort a list of n numbers into ascending order.

(ii) Write down the circumstances under which a bubble sort stops.

(4)

A bubble sort, starting at the left-hand end of the list, is used to sort a list of ten numbers into ascending order. After a number of passes the list reads

0.9 1.2 1.5 0.5 1.4 1.1 0.7 1.7 2.2 3.2

(b) Determine the maximum number of passes that could have taken place on this list. You must give a reason for your answer.

(2)

(c) Complete the bubble sort to produce a list of the numbers in ascending order. You only need to give the state of the list after each complete pass.

(4)

(d) Use the first-fit decreasing bin packing algorithm to determine how the ten numbers listed above can be packed into bins of size 4

(3)

(Total 13 marks)

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2.

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Question 2 continued

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Question 2 continued

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Question 2 continued

(Total 13 marks)

Q2

Write your answers in the D1 answer book for this paper.

1.

16 23 18 9 4 20 35 5 17 13 6 11

The numbers in the list represent the weights, in kilograms, of twelve parcels. The parcels are to be transported in containers that will each hold a maximum weight of 45 kg.

(a) Calculate a lower bound for the number of containers needed. You must make your method clear.

(2)

(b) Use the first-fit bin packing algorithm to allocate the parcels to the containers.

(3)

(c) Carry out a bubble sort, starting at the left-hand end of the list, to produce a list of the weights in descending order. You should only give the state of the list after each pass.

(4)

(d) Use the first-fit decreasing bin packing algorithm to allocate the parcels to the containers.

(3)

(Total 12 marks)

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1.

16 23 18 9 4 20 35 5 17 13 6 11

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Question 1 continued

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Question 1 continued

(Total 12 marks)

Q1

3.

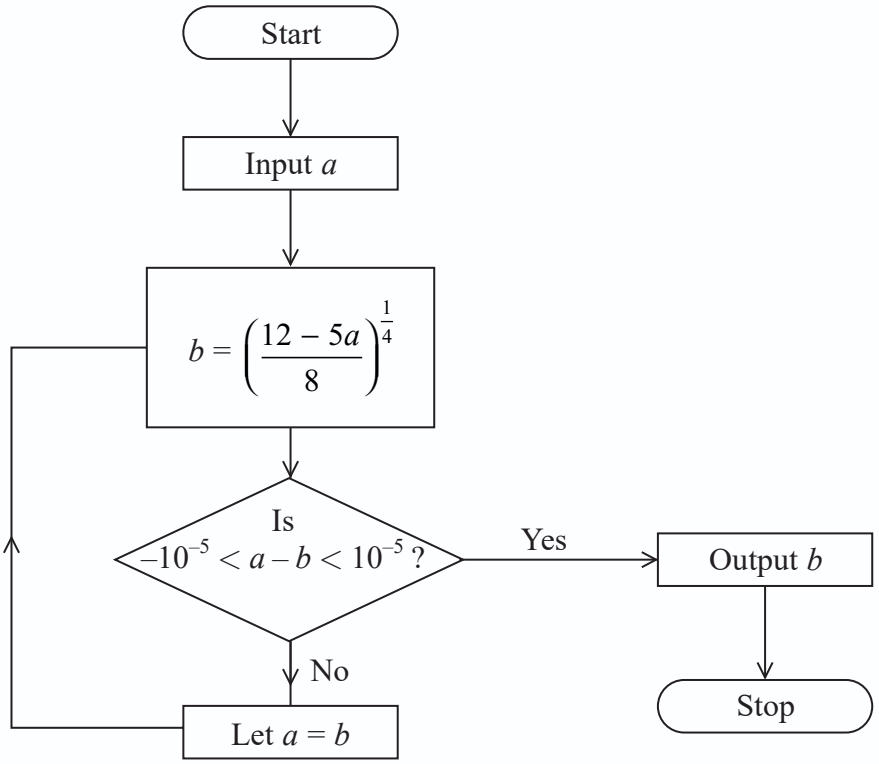


Figure 2

An algorithm for finding the positive real root of the equation $8x^4 + 5x - 12 = 0$ is described by the flow chart shown in Figure 2.

(a) Use the flow chart, with $a = 1$, to complete the table in the answer book, stating values to at least 6 decimal places. Give the final output correct to 5 decimal places.

(4)

Given that the value of the input a is a non-negative real number,

(b) determine the set of values for a that **cannot** be used to find the positive real root of $8x^4 + 5x - 12 = 0$ using this flow chart.

(2)

(Total 6 marks)

Leave blank

3. *You may not need to use all the rows in this table.*
You should write your final output in the box provided at the bottom of this page.

a	b	Is $-10^{-5} < a - b < 10^{-5}$?

Final output	

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Question 3 continued

Q3

(Total 6 marks)

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Write your answers in the D1 answer book for this paper.

1. 175 135 210 105 100 150 60 20 70 125

The numbers in the list above represent the weights, in kilograms, of ten crates. The crates are to be transported in trucks that can each hold a maximum total crate weight of 300 kg.

(a) Calculate a lower bound for the number of trucks that will be needed to transport the crates.

(2)

(b) Using the list provided, carry out a bubble sort to produce a list of the weights in descending order. You need only give the state of the list after each pass.

(4)

(c) Use the first-fit decreasing bin packing algorithm to allocate the crates to the trucks.

(3)

(Total for Question 1 is 9 marks)

1.

175 135 210 105 100 150 60 20 70 125

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Question 1 continued

(Total for Question 1 is 9 marks)

2. A list of eleven numbers is to be sorted into descending order.

After one pass, the quick sort algorithm produces the following list

17 33 14 25 23 28 21 13 9 6 10

(a) State, with a reason, which number was used as a pivot for the first pass.

(1)

(b) Starting at the left-hand end of the above list, obtain the fully sorted list using a bubble sort. You need to write down only the list that results at the end of each pass.

(3)

(c) Apply the first-fit decreasing bin packing algorithm to the fully sorted list to pack the numbers into bins of size 85

(2)

(Total for Question 2 is 6 marks)

2. 17 33 14 25 23 28 21 13 9 6 10

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Question 2 continued

(Total for Question 2 is 6 marks)

3. In this question, the function INT(X) is the largest integer less than or equal to X.

For example,

$$\text{INT}(5.7) = 5$$

$$\text{INT}(8) = 8$$

$$\text{INT}(-2.3) = -3$$

Consider the following algorithm.

Step 1 Input N

Step 2 Calculate $A = N \div 10$

Step 3 Let $B = \text{INT}(A)$

Step 4 Calculate $C = B \times 10$

Step 5 Calculate $D = N - C$

Step 6 Output D

Step 7 Replace N by B

Step 8 If $N = 0$ then STOP, otherwise go back to Step 2

(a) Complete the table in the answer book, using $N = 4217$, to show the results obtained at each step of the algorithm.

(4)

(b) Explain how the output values of the algorithm relate to the original input N, where N is any positive integer.

(2)

(Total for Question 3 is 6 marks)

3. *You may not need to use all the rows in this table.*

It may not be necessary to complete all boxes in each row.

N	*A*	*B*	*C*	*D*

Outputs: _____

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Question 3 continued

(Total for Question 3 is 6 marks)

Write your answers in the D1 answer book for this paper.

1.

5.2 4.7 6.5 4.5 3.1 5.1 1.8 2.9 3.4 3.8 1.2

(a) Use the first-fit bin packing algorithm to determine how the eleven numbers listed above can be packed into bins of size 14

(3)

(b) The list of numbers is to be sorted into **ascending** order. Use a quick sort to obtain the sorted list. You should show the result of each pass and identify your pivots clearly.

(4)

(c) Apply the first-fit decreasing bin packing algorithm to the sorted list to pack the numbers into bins of size 14

(3)

(d) Explain why the number of bins used in part (c) is optimal.

(1)

(e) Use the binary search algorithm to **try** to locate 3.0 in the list of numbers. Clearly indicate how you choose your pivots and which part of the list is rejected at each stage.

(3)

(Total for Question 1 is 14 marks)

1.

5.2 4.7 6.5 4.5 3.1 5.1 1.8 2.9 3.4 3.8 1.2

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Question 1 continued

Question 1 continued

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Question 1 continued

(Total for Question 1 is 14 marks)

7. The numbers listed below are to be packed into bins of size n, where n is a positive integer.

14 20 23 17 15 22 19 25 13 28 32

A lower bound for the number of bins required is 4

(a) Determine the range of possible values of n. You must make your method clear.

(3)

(b) Carry out a quick sort to produce a list of the numbers in descending order. You should show the result of each pass and identify your pivots clearly.

(4)

When the first-fit bin packing algorithm is applied to the **original** list of numbers, the following allocation is achieved.

Bin 1: 14 20 23 15

Bin 2: 17 22 19 13

Bin 3: 25 28

Bin 4: 32

When the first-fit decreasing bin packing algorithm is applied to the sorted list of numbers, the following allocation is achieved.

Bin 1: 32 28

Bin 2: 25 23 22

Bin 3: 20 19 17 15

Bin 4: 14 13

(c) Determine the value of n. You must explain your reasoning fully.

(3)

(Total 10 marks)

TOTAL FOR PAPER: 75 MARKS

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7.

14 20 23 17 15 22 19 25 13 28 32

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Question 7 continued

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Question 7 continued

Q7

(Total 10 marks)

TOTAL FOR PAPER: 75 MARKS

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